

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 1

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(Story Proof) Define $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ as the number of ways to partition $\{1, 2, \dots, n\}$ into k non-empty subsets, or the number of ways to have n students split up into k groups such that each group has at least one student. For example, $\left\{ \begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right\} = 7$ because we have the following possibilities:

- $\{1\}, \{2, 3, 4\}$
- $\{1, 2\}, \{3, 4\}$
- $\{2\}, \{1, 3, 4\}$
- $\{1, 3\}, \{2, 4\}$
- $\{3\}, \{1, 2, 4\}$
- $\{1, 4\}, \{2, 3\}$
- $\{4\}, \{1, 2, 3\}$

Prove the following identities:

(a)

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}.$$

Hint: I'm either in a group by myself or I'm not.

(b)

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}.$$

Hint: First decide how many people are not going to be in my group.

Solution

(a) $\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\}$ has two situations:

- (i) I am alone in one group. The remaining n people are in $k-1$ groups, giving $\left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\}$ ways.
- (ii) I am part of a group. The n people are in k groups, and I have k options, giving $k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ways.

Therefore,

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}.$$

(b) $\binom{n}{j}$ means choosing j people from n people. To partition $n+1$ people into $k+1$ groups, assume j people are not going to be in my group. They should be in k groups, giving $\left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$ ways. For a particular j , we need $j \geq k$. Thus all possible choices give

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}.$$

Problem 2

A *norepeatword* is a sequence of at least one (and possibly all) of the usual 26 letters a, b, c, ..., z, with repetitions not allowed. For example, "course" is a norepeatword, but "statistics" is not. Order matters, e.g., "course" is not the same as "source". A norepeatword is chosen randomly, with all norepeatwords equally likely. Show that the probability that it uses all 26 letters is very close to $1/e$.

Solution

The number of norepeatwords with k letters is

$$26 \cdot 25 \cdots (26 - k + 1) = \frac{26!}{(26 - k)!} \quad (\text{order matters}).$$

The number with 26 letters is $26!$, and the total number of norepeatwords is

$$\sum_{k=1}^{26} \frac{26!}{(26 - k)!}.$$

Therefore,

$$P = \frac{26!}{\sum_{k=1}^{26} \frac{26!}{(26 - k)!}} = \frac{1}{\sum_{j=0}^{25} \frac{1}{j!}}.$$

We know that

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots = \sum_{j=0}^{\infty} \frac{1}{j!}.$$

Since

$$e - \sum_{j=0}^{25} \frac{1}{j!} = \frac{1}{26!} + \frac{1}{27!} + \cdots$$

is very small, $\sum_{j=0}^{25} 1/j! \approx e$, and hence $P \approx 1/e$.

Problem 3

Given $n \geq 2$ numbers $(a_1, a_2, \dots, a_n)$ with no repetitions, a bootstrap sample is a sequence $(x_1, x_2, \dots, x_n)$ formed from the a_j 's by sampling with replacement with equal probabilities. Bootstrap samples arise in a widely used statistical method known as the bootstrap. For example, if $n = 2$ and $(a_1, a_2) = (3, 1)$, then the possible bootstrap samples are $(3, 3)$, $(3, 1)$, $(1, 3)$, and $(1, 1)$.

- How many possible bootstrap samples are there for $(a_1, \dots, a_n)$?
- How many possible bootstrap samples are there for $(a_1, \dots, a_n)$, if order does not matter (in the sense that it only matters how many times each a_j was chosen, not the order in which they were chosen)?
- One random bootstrap sample is chosen (by sampling from $a_1, \dots, a_n$ with replacement, as described above). Show that not all unordered bootstrap samples (in the sense of (b)) are equally likely. Find an unordered bootstrap sample $\mathbf{b}_1$ that is as likely as possible, and an unordered bootstrap sample $\mathbf{b}_2$ that is as unlikely as possible. Let p_1 be the probability of getting $\mathbf{b}_1$ and p_2 be the probability of getting $\mathbf{b}_2$ (so p_i is the probability of getting the specific unordered bootstrap sample $\mathbf{b}_i$). What is p_1/p_2 ? What is the ratio of the probability of getting an unordered bootstrap sample whose probability is p_1 to the probability of getting an unordered sample whose probability is p_2 ?

Solution

- Each place in $(x_1, x_2, \dots, x_n)$ has n options, so the number of samples is

$$n \times n \times \dots \times n = n^n.$$

- Order does not matter. Let c_i be how many times a_i is chosen. Then

$$c_1 + c_2 + \dots + c_n = n, \quad c_i \geq 0.$$

Placing $n - 1$ bars among $2n - 1$ positions gives

$$\binom{2n-1}{n-1}.$$

- Take $n = 3$ as an example. The unordered bootstrap samples have three situations:

- No repeats: $\{a_1, a_2, a_3\}$ corresponds to $3! = 6$ ordered sequences and has probability $6/27 = 2/9$.
- One element repeats: for example, $\{a_1, a_1, a_2\}$ or $\{a_1, a_1, a_3\}$. Each corresponds to 3 ordered sequences and has probability $3/27 = 1/9$.
- Three repeats: for example, $\{a_1, a_1, a_1\}$ has probability $1/27$.

They are not equally likely.

To find the most likely $\mathbf{b}_1$, each element should be sampled, and

$$p_1 = \frac{n!}{n^n}.$$

To find the least likely $\mathbf{b}_2$, only one element is chosen, and

$$p_2 = \frac{1}{n^n}.$$

Thus

$$\frac{p_1}{p_2} = n!.$$

Only $\{a_1, \dots, a_n\}$ has probability p_1 , so $P_1 = p_1$.

The n unordered samples with probability p_2 are $\{a_1, \dots, a_1\}, \dots, \{a_n, \dots, a_n\}$. Thus $P_2 = np_2$, and the required ratio is

$$\frac{P_1}{P_2} = \frac{n!}{n} = (n-1)!.$$

Problem 4

In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $\mathbf{p} = (p_1, p_2, \dots, p_{365})$ be the vector of birthday probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers). The k th elementary symmetric polynomial in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}.$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables.

For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$. Now let $k \geq 2$ be the number of people.

(a) Find a simple expression for the probability that there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.

(b) Explain intuitively why it makes sense that P (at least one birthday match) is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.

(c) The famous arithmetic mean-geometric mean inequality says that for $x, y \geq 0$

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x-y)^2 \geq 0$. Define $\mathbf{r} = (r_1, \dots, r_{365})$ by $r_1 = r_2 = (p_1 + p_2)/2$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$$e_k(x_1, \dots, x_n) = x_1x_2e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$$

show that

$$P(\text{at least one birthday match} \mid \mathbf{p}) \geq P(\text{at least one birthday match} \mid \mathbf{r})$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the given $\mathbf{r}$ notation means that the birthday probabilities are given by $\mathbf{r}$. Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

Solution

(a) The probability that the k people all have different birthdays is

$$k! \sum_{1 \leq j_1 < \dots < j_k \leq 365} p_{j_1} \dots p_{j_k} = k!e_k(\mathbf{p}).$$

So the probability of at least one birthday match is

$$P(\text{match} \mid \mathbf{p}) = 1 - k!e_k(\mathbf{p}).$$

(b) If there are m days available and only two people, then

$$P(\text{match}) = \sum_{j=1}^m p_j^2.$$

We know that

$$\sum_{j=1}^m \left(p_j - \frac{1}{m}\right)^2 = \sum_{j=1}^m p_j^2 + \frac{1}{m} - \frac{2}{m} \sum_{j=1}^m p_j = \sum_{j=1}^m p_j^2 - \frac{1}{m} \geq 0.$$

Thus $\sum_{j=1}^m p_j^2 \geq 1/m$, with equality when $p_1 = \dots = p_m = 1/m$. So, intuitively, $P(\text{match})$ is minimized when $p_j = 1/365$ for all j .

(c) To verify the equation, for $e_k(x_1, \dots, x_n)$ we have three situations:

- (i) x_1, x_2 are in the product, giving $x_1 x_2 e_{k-2}(x_3, \dots, x_n)$;
- (ii) x_1 or x_2 is in the product, giving $(x_1 + x_2) e_{k-1}(x_3, \dots, x_n)$;
- (iii) neither is in the product, giving $e_k(x_3, \dots, x_n)$.

Hence

$$e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n).$$

Let $\mathbf{r} = (r_1, \dots, r_{365})$, where $r_1 = r_2 = (p_1 + p_2)/2$ and $r_j = p_j$ for $3 \leq j \leq 365$. Since

$$P(\text{match} \mid \mathbf{p}) = 1 - k! e_k(\mathbf{p}),$$

we have

$$P(\text{match} \mid \mathbf{p}) - P(\text{match} \mid \mathbf{r}) = k! [e_k(\mathbf{r}) - e_k(\mathbf{p})].$$

Put $\mathbf{x} = (p_3, \dots, p_{365})$. Since $r_j = p_j$ for $j \geq 3$,

$$\begin{aligned} e_k(\mathbf{r}) - e_k(\mathbf{p}) &= (r_1 r_2 - p_1 p_2) e_{k-2}(\mathbf{x}) + (r_1 + r_2 - p_1 - p_2) e_{k-1}(\mathbf{x}) \\ &= \frac{(p_1 - p_2)^2}{4} e_{k-2}(\mathbf{x}) \geq 0. \end{aligned}$$

Therefore $P(\text{match} \mid \mathbf{p}) \geq P(\text{match} \mid \mathbf{r})$, and if $\mathbf{p} \neq \mathbf{r}$, it is strictly unequal.

Given an arbitrary vector $\mathbf{p} = (p_1, \dots, p_{365})$ with $p_1 + \dots + p_{365} = 1$, to minimize the probability, take

$$p'_1 = p'_2 = \frac{p_1 + p_2}{2}, \quad p''_1 = p''_3 = \frac{p'_1 + p_3}{2}, \quad \dots$$

After a large number of iterations, the final minimizing vector has $p_j = 1/365$ for all j .